

Rice Leaf Disease Detection Using Convolutional Neural Networks

Dr.Raghavendar Raju L¹
Assistant Professor, Dept of CSE
Matrusri Engineering College
Hyderabad, India
lraghavendarraju@matrusri.edu.in

Navateja G²
UG Scholar, Dept of CSE
Matrusri Engineering College
Hyderabad, India
gogulanavateja1910@gmail.com

Abstract— Agriculture stands as a significant revenue-generating sector and a primary source of livelihood in India. In one of India's largest regions, rice cultivation is pivotal for food production. However, rice crops are notably susceptible to diseases, causing substantial losses in the agricultural sector. Plant pathologists are actively seeking precise and dependable methods for diagnosing diseases in rice plants. Machine learning has proven effective in various aspects of crop remote sensing, especially in the classification of crop diseases. Currently, deep learning is a prominent research focus for identifying crop diseases. In this study, we've developed an efficient method for detecting diseases in rice plants using a convolutional neural network (CNN) approach. This paper primarily addresses three well-known rice diseases: leaf Blast and Brown Spot, both caused by fungi, and bacterial leaf blight caused by bacteria. The article introduces an effective approach for recognizing and identifying rice plant diseases based on the size, shape, and color of lesions in leaf images. To detect the three rice diseases, the method employs a fully connected CNN model that was trained using 2,000 image samples of each diseased type and 2,000 image samples of healthy rice leaves. This research represents a promising step in automating the diagnosis and identification of rice plant diseases, offering potential benefits to the agricultural sector. The analysis of the results indicates that the proposed fully connected Convolutional Neural Network (CNN) method is both rapid and highly effective. It achieves an impressive accuracy rate of 99.7% on the dataset. This level of accuracy is notably superior to that of existing plant disease recognition and classification methods.

Keywords— Convolutional Neural Networks (CNNs), CNN Architecture, Machine Learning, Image Classification, Deep Learning.

I. INTRODUCTION

The agricultural sector is the lifeblood of India, supporting around 70% of the population and serving as a cornerstone of the nation's economy. Among the various crops cultivated, rice holds a place of paramount importance, serving as a staple food for a significant portion

of the population. The entire Indian economy is intricately linked with the production of rice and its associated agro-industrial products. The successful growth and development of rice crops depend on a multitude of factors, including soil quality, climate conditions, irrigation, topography, seed selection, and various biological factors. Among these, plant diseases stand out as a critical biological factor that can lead to substantial reductions in rice production. These diseases, primarily caused by bacteria and fungi, have the potential to cause significant harm. In India, they are responsible for a substantial 15-20% loss in rice production.

Notable infections like brown spot shown in Fig.1, bacterial leaf blight shown in Fig.2, and leaf blast shown in Fig.3 are well-documented as major culprits behind crop damage and declining yields. These diseases pose a significant challenge to the agricultural sector, demanding innovative solutions to safeguard this vital crop and, by extension, the nation's food security and economic well-being. Detection of these diseases is not only essential for protecting food security but also for ensuring the economic stability of the agricultural industry.



Fig.1 Rice leaf with brown spot disease

Traditional methods of disease detection in these crops have relied heavily on visual inspection by trained agronomists or farmers, often leading to delayed diagnoses and inconsistent results. The need for more efficient, accurate, and scalable solutions has driven research into the integration of cutting-edge technology into agriculture.



Fig.2 Rice leaf with bacterial leaf blight disease



Fig.3 Rice leaf with blast disease

This research embarks on a mission to revolutionize crop management, offering a lifeline to farmers and agriculture experts in the form of early, precise, and actionable disease detection. A sophisticated model deeply rooted in Convolutional Neural Networks (CNNs). This model isn't just a harbinger of disease; it possesses the remarkable capability to identify specific diseases in rice plants and, significantly, to declare a leaf healthy. The importance of this endeavor cannot be overstated. Early disease detection, as enabled by CNNs, becomes the linchpin of a new era in agriculture. As we delve into the fabric of this research, the essential steps unravel – from data collection and preprocessing to the meticulous architecture design of our model, its training and evaluation, to its practical deployment.

datasets. Subsequent to image acquisition, the pivotal step is image pre-processing. This stage encompasses diverse operations aimed at refining and readying the dataset for analysis. These operations encompass activities such as image resizing, background removal, noise reduction, and augmentation, among others.

After acquiring the images, the next step is image segmentation, which serves the purpose of isolating the infected regions within the images. This process is vital for pinpointing the relevant features based on the specific requirements of the analysis. Following segmentation, the feature extraction phase is employed to identify the most distinguishing attributes of the selected objects in the images. Historically, image processing techniques have been used for this purpose. However, modern approaches increasingly rely on deep neural networks for automatic feature extraction from images. Typically, this involves extracting texture, shape, and color-related characteristics from regions of interest (ROI) in the images. The subsequent classification technique is then applied to categorize the given image sample into a specific output class.

This research isn't an isolated advancement; it's a vital contribution to agriculture's arsenal. It places in the hands of farmers and experts a tool that empowers them with the earliest possible data about the health of their rice crops. Beyond disease diagnosis, this technology paves the way for timely interventions, efficient resource allocation, and ultimately, a bolstering of global food security.

II. LITERATURE REVIEW

The traditional method of detecting and analysing diseases in plant leaves are manual. This method of detecting disease in plant is very costly, time consuming and subjective [1]. In the past few years, there have been significant developments in the field of plant disease recognition and classification. Researchers and computer scientists have successfully built models for the automatic detection of plant diseases by utilizing key image features such as texture, color, shape, and size. These computer-based disease detection methods have effectively addressed the limitations associated with manual disease recognition approaches [2, 3]. We conducted an extensive literature survey on related methods proposed in duration from 2007 to 2021. Literature review of some of relevant articles are discussed in this section.

Huang [4] proposed a method for detecting plant diseases in Phalaenopsis orchids. The approach involved a well-structured segmentation process comprising four key stages. Initially, a Bayes classifier was employed to eliminate the plant's vessel. Subsequently, an exponential transformation was applied to equalize the image,

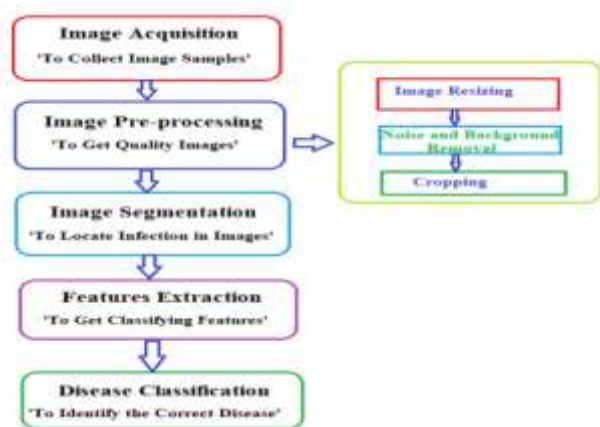


Fig.4 The general process of Leaf disease classification using machine learning

The initial phase of any image processing application involves image acquisition, where the goal is to collect images of the afflicted plant sections. These images are sourced either from the field or established benchmark datasets and serve as the basis for training and testing

facilitating the identification of infected areas through rough computation. Further refinement was achieved by equalizing the estimated infected regions. Texture and color features were then extracted from the segmented image using a gray level co-occurrence matrix. Finally, these extracted features were utilized as input for a multi-layer perceptron, enabling accurate disease classification and identification within the plant.

Yao et al. [5] introduced a classification model based on support vector machines for the identification of three different diseases affecting rice plants. Their approach involved a color transformation to obtain a 2-channel image from the original RGB input image. Infected regions were precisely determined using Otsu's image segmentation method. Subsequently, color, texture, and shape features were extracted from these identified infected areas. The support vector machine classifier was then trained using these features, enabling the model to make accurate classifications of the diseases.

Verma and Dubey [6] introduced an innovative approach for the detection of rice plant diseases using video image processing. They employed two deep learning methods: the Simple Recurrent Neural Network (SRNN) and Long Short-Term Memory (LSTM). The process involved capturing videos of rice plants using a mobile device and dividing them into two sets for testing and validation. To segment these videos, they utilized linear color space transformation and color indexing techniques. Following video segmentation, features such as standard deviation, entropy, and fuzzy attributes were extracted as time-series datasets. These datasets were employed to train the LSTM-SRNN-based model, enabling it to distinguish between diseased and healthy rice plants. Remarkably, the authors reported an impressive 97.99% prediction accuracy for their proposed model. Additionally, their method outlined ways to adapt learning dynamically.

Prajapati et al. [7] developed a predictive model based on Support Vector Machines (SVM) for the identification of diseases in paddy crops. Their approach involved using images of both diseased and healthy leaves. To enhance image quality, they applied image processing techniques. Subsequently, machine learning algorithms were employed to construct the predictive model.

The image pre-processing phase focused on removing background noise from the images, improving their overall quality. K-means clustering was utilized to isolate the diseased portions of the leaves. Features such as texture, color, and shape of the diseased leaves were extracted for disease prediction. The results were promising, with the proposed technique achieving an accuracy of 93.33% on the training dataset and 73.33% on the test dataset. These findings suggest the model's effectiveness in distinguishing between healthy and diseased paddy leaves.

III. METHODOLOGY

3.1 Data Collection and Preprocessing:

Data Acquisition: Our research involved two experiments that relied on the Rice Leaf dataset obtained from Kaggle. This dataset is composed of four categories of rice leaves, each category containing 2,000 images, as outlined in Table 1. These categories encompass both healthy and unhealthy rice leaves, with the latter category including three specific diseases: brown spot, bacterial leaf blight, and blast. Figure 4 displays examples of rice leaf images, both healthy and diseased, which were used in our study.

Table 1 Dataset description

Leaves class	Number of images
Healthy	2000
Bacterial leaf blight	2000
Blast	2000
Brown spot	2000



(a)Diseased Leaf Samples



(b)Healthy Leaf Samples

Fig.5 Dataset samples of rice leaf images **a** diseased leaf samples, **b** healthy leaf samples

Data Labeling: To prepare the dataset for training Convolutional Neural Networks (CNNs) accurately, it was necessary to annotate each image. This process involved categorizing the plant images as either "healthy" or specifying the type of disease present, such as "Brown Spot," "Bacterial Leaf Blight," "Blast,". These annotations are crucial for training the CNNs to effectively identify and classify various plant diseases.

Data Splitting: The dataset was partitioned into distinct subsets: a training set, a validation set, and a test set. This division was meticulously executed to guarantee diversity and representativeness within each subset. The training set

was employed to train the models, the validation set to fine-tune and optimize them, and the test set to evaluate their overall performance. This careful partitioning process was essential for producing reliable and generalizable results.

3.2 Data Augmentation:

Implemented data augmentation techniques to increase the dataset's diversity and improve model generalization. Augmentation methods may include random flips, rotations, zooming, and brightness adjustments.

3.3 CNN Architecture Design:

A Convolutional Neural Network (CNN) is a specialized type of neural network designed for image processing tasks. In the context of our research, the CNN model plays a pivotal role in accurately identifying and classifying plant diseases while also providing potential remedies. This CNN architecture, as depicted in Figure 5, comprises the following essential layers:

Input Layer: Serving as the initial point of entry, the Input Layer accepts preprocessed grayscale images. These images have been standardized in size, typically resized to dimensions like 256x256 pixels for consistency.

Convolutional Layers: Convolution layers are at the core of any CNN. They employ learnable filters to scan input images, detecting various spatial features. In our model, these layers play a crucial role in capturing meaningful features from the input images. These features can range from basic elements like edges to more complex patterns associated with plant diseases.

Rectified Linear Unit (ReLU) Layer: Following convolution, the ReLU activation function is applied. ReLU introduces non-linearity by setting negative values to zero and leaving positive values unchanged. This non-linearity is pivotal as it enables the model to learn intricate patterns and relationships within the data, thereby enhancing its feature representation capabilities.

Pooling Layer: Pooling layers immediately succeed the ReLU layer. Their primary purpose is to downsample spatial dimensions, reduce overfitting, and optimize computational efficiency. Common pooling methods include max-pooling, which selects the maximum value from a group of values, and average-pooling, which calculates the mean value. These techniques help retain vital information.

Fully Connected Layers: The Fully Connected Layer, a fundamental component, connects every neuron to all neurons in the previous and subsequent layers. This layer is instrumental in the classification process. In our model, the fully connected layers process the high-level features extracted from previous layers, facilitating disease detection and classification. The output layer of these fully connected layers delivers the ultimate prediction, identifying plant diseases and, notably, suggesting remedies based on the classification.

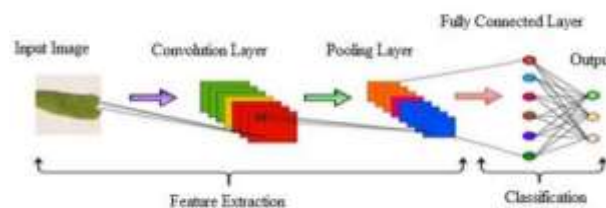


Fig.6 Fundamental Architecture of CNN

Output Layer: The Output Layer is where the model provides its final determination. It offers insight into whether a plant is healthy or afflicted by a specific disease. Additionally, this layer can generate recommendations or remedies based on the disease classification, making it a valuable resource for maintaining plant health.

3.4 Model Training:

Trained the CNN model on the training dataset using a loss function, such as categorical cross-entropy, is selected to measure the error between predicted and actual disease classes and Optimization algorithms like Adam or Stochastic Gradient Descent (SGD) are chosen to minimize this loss function.

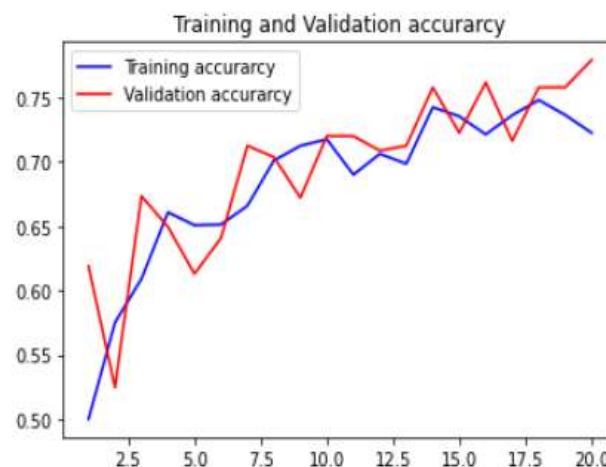
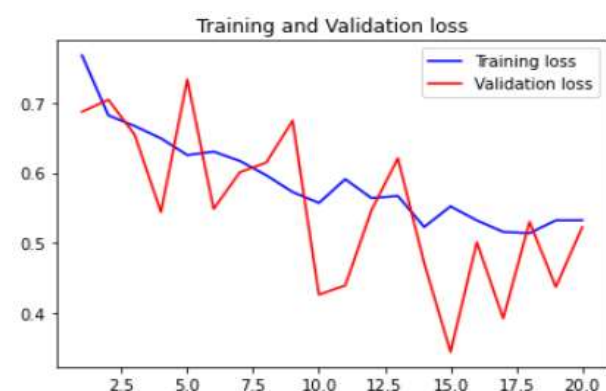


Fig.7 Training and validation accuracy and loss



3.5 Implementation

The implementation of rice leaf disease detection using Convolutional Neural Networks (CNN) algorithm is a crucial step in transforming the theoretical concepts into a practical and deployable system. This stage involves translating the design and architecture of the CNN model into actual code, training the model on a dataset of rice leaf images, and creating a user-friendly interface for farmers or agricultural experts to interact with the system.

Data Collection: Collected a diverse and well-annotated dataset of rice leaf images containing healthy leaves and leaves affected by various diseases. Ensure that the dataset covers different disease severities and includes representative samples of different rice leaf varieties.

Data Preprocessing: Preprocess the collected images to ensure uniformity in size and format. Perform data augmentation techniques like rotation, flipping, and zooming to increase the dataset size and improve the model's generalization.

Split Dataset: Divide the dataset into training, validation, and test sets. The training set will be used to train the CNN model, the validation set for hyperparameter tuning and performance evaluation, and the test set to assess the model's generalization.

Building the CNN Model: Designed the architecture of the CNN model tailored for rice leaf disease detection. Created convolutional layers for feature extraction, activation functions for non-linearity, pooling layers for down-sampling, and fully connected layers for classification.



Fig.8. System Architecture.

CNN Model Training: Training the CNN model using the training dataset. Utilize a suitable optimization algorithm and an appropriate loss function to minimize the difference between predicted and actual disease labels.

Validation and Model Evaluation: Evaluating the trained CNN model using the validation dataset. Calculate relevant metrics such as accuracy, precision, recall, and F1-score to assess the model's performance and fine-tune further if necessary.

Model Testing: Evaluating the final trained model using the test dataset to ensure its performance on unseen data and validate its ability to detect rice leaf diseases accurately.

User Interface Development: Created a user-friendly interface with the help of React JS and FastAPI that allows farmers or agricultural experts to upload images of rice leaves for disease detection. The interface provides a clear and interpretable results, indicating the detected disease and its severity.

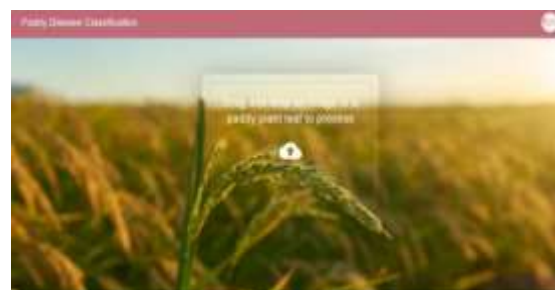


Fig.9 User Interface of Website

System Integration and Deployment: Finally we integrate the trained CNN model and the user interface into a complete system. Which helps the user to upload the images very easily and easy to get the results.

IV. RESULTS

To assess the efficacy of our leaf disease classification model, we must conduct a comprehensive analysis of its predictive accuracy when applied to the dataset's images. This evaluation is instrumental in determining the model's competence in recognizing and categorizing the diverse diseases that afflict rice leaves. The outcomes derived from the model's predictions on a variety of leaf images are pivotal. They furnish us with a measure of how proficiently the model can differentiate between healthy rice leaves and those afflicted by diseases like bacterial leaf blight, blast, and brown spot. The accuracy of these predictions serves as a critical yardstick for assessing the model's performance, particularly in real-world scenarios. This assessment is indispensable for the model's practical application in the identification and mitigation of rice leaf diseases.



Fig.10. Predicted as Bacterial Leaf Blight with 100% Accuracy

V. CONCLUSION AND FUTURE WORK

Farmers are currently grappling with the challenge of identifying diseases in their rice leaves and finding effective solutions in the form of pesticides or insecticides. To tackle this issue, we've harnessed the potential of the Convolutional Neural Network (CNN) algorithm to create a machine learning model. This model is designed to accurately detect common problems such as Rice Blast, Bacterial Leaf Blight, Brown spot, and identify healthy rice leaves.

However, our solution goes beyond mere identification. It provides tailored recommendations for the application of insecticides or pesticides to effectively control Rice Blast and Bacterial Leaf Blight. What sets our system apart is its robustness, user-friendliness, speed, and cost-effectiveness when compared to existing methods.

In our vision for the future, we plan to broaden the scope of this technology to cover a wider range of rice leaf diseases. Moreover, we are working on developing a mobile application, which will be available for free on platforms like Google Play. This app will not only include support for additional rice leaf diseases but will also integrate with models for various other crop types, allowing the system to predict diseases in a wide range of crops.

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